

Object Oriented Programming

Chapter One: Introduction to Object-Oriented Programming

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- Basic Concepts
 - ▶ Introduction to programming
 - ▶ Basics of programming
 - ▶ Overview of OO principles
 - ▶ Overview of Java Programming
 - ▶ Types of Java Program
 - ▶ Java Application

Introduction to programming

What is Computer?

- **Computer** is an electronic device that **accepts data**, **performs** computations, and **makes logical decisions** according to instructions that have been given to it;
 - ▶ Then produces meaningful information in a form that is useful to the user.
- A computer is an electronic device that **manipulates information, or data**.
- It has the ability to **store**, **retrieve**, and **process** data.

Introduction to programming

Definition

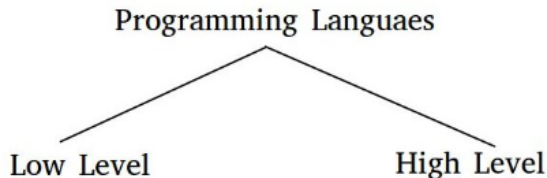
- **Computer Programs**, Software Programs, or just Programs is instructions that tells the computer **what to do**.
- Computer programming is the process that professionals use to write code that instructs how a computer, application or software program performs.
- **Writing computer programs** means **writing instructions**
 - ▶ That will make the computer follow and run a program **based on those instructions**.

Introduction to programming

- Computer programs (also **known** as **source code**) is often written by **professionals known as Computer Programmers** (simply programmers).
- Source code is written in **one of programming languages**.
- **A programming language** is an **artificial language** that can be used to control the behavior of a machine, particularly a computer.
- Programming languages, **like natural language** (such as Amharic), are defined by **Syntax and semantic rules**
 - ▶ **Syntax**: is the set of **predefined rules, processes, and protocols** that **everyone should follow**, if they want an error-free code.
 - ▶ **Semantic**: is the **meaning** of the **expressions, statements, and program units**.

Introduction to programming

- Programming languages can be divided in to two major categories:



Programming languages categories

Low-level Languages

- Computers only understand one language and that is binary language or the language of 1s and 0s.
- All the instructions were given in binary form.
- Although the computer easily understood these programs,
 - ▶ It proved too **difficult** for a **normal human being to remember** all the instructions **in the form of 0s and 1s**.
 - ▶ **Low-level Languages** is a type of language, which is **machine friendly**.
 - ▶ Two common types of low-level programming languages are **assembly language** and **machine language**.

Low-level Languages Type

- **Machine-level language**

- ▶ **The Machine-level language** is a language that consists of a set of instructions that are in the binary form **0 or 1**.
 - ★ So, the instructions **given to the computer can be only in binary codes**
- ▶ If we write a program in one computer will **no longer be valid in another computer**.
- ▶ **It does not require any translator** as the machine code is **directly executed by the computer**.
- ▶ It is a **first-generation** programming language.

Low-level Languages Type

- **Assembly language**

- ▶ **Assembly language** correspondences **symbolic instructions** and **executable machine codes**
- ▶ Since assembly language instructions are written in English words like **mov, add, sub**, so it is easier to write and understand.
- ▶ As we know that computers can only understand the machine-level instructions
 - ★ So we require a translator that **converts the assembly code into machine code**.
- ▶ The translator used for translating the code is known as an **assembler**.

Programming languages categories

High-level languages

- A high-level language is any programming language that enables development of a program in a **much more user-friendly programming context**
- High-level language is generally **independent of the computer's hardware architecture.**
- High-level language **does not require addressing hardware constraints when developing a program.**
- High-level languages also **require translation to machine language before execution.**

Programming languages categories

High-level languages

- This translation is accomplished by either a compiler or an interpreter.
 - ▶ **Compiler**
 - ★ Compilers translate the **entire source code program before execution.**
 - ▶ **Interpreters**
 - ★ Interpreters translate source code programs **one line at a time.**

What is Java?

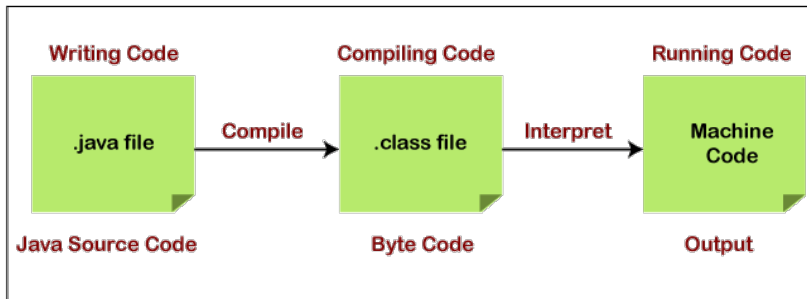
- Java is a **programming language**.
- java is a **high level, robust, object-oriented** and **secure programming language**.
- Java is a powerful **general-purpose programming language**.
 - ▶ It is used to develop **desktop and mobile applications, big data processing**, and so on.
- Java is **one of the most popular programming languages**.
 - ▶ Because, according to Oracle, **Java runs on 3 billion devices worldwide**

What is Java?

- The Java language is **object-oriented** and **programs written in Java are portable**
 - ▶ That is, they can be run on many **different platforms**
- Java is **different from other programming languages** because a **Java program is compiled and interpreted**
 - ▶ In many other languages, the program is **either compiled or interpreted**.
- The Java **compiler translates** the program into an intermediate language called **Java bytecodes**.

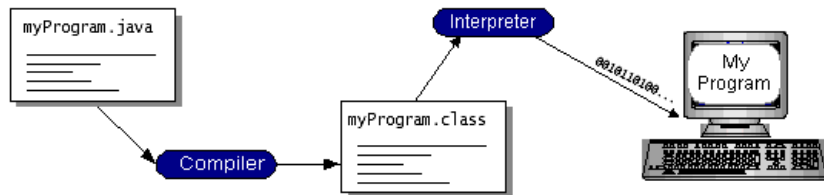
Introduction to Java

- a java program is **compiled**, **java bytecode** is generated.
- **java bytecode** is the **machine code** in the form of a **.class file**.
- Java bytecodes are **platform-independent**
 - ▶ This means they can be run on different operating systems



Introduction to Java

- bytecode



What is Java?

- **Compilation only** needs to happen **one time**, whereas **interpretation** happens **every time the program is executed**.
- The source code for a Java program is saved to **one or more .java files**.
 - ▶ The compiler generates a **.class file for each .java file**
- **A .java file can be opened and viewed in any text editor**
 - ▶ But a .class file is bytecode **so it cannot be viewed in the same way**

Why Java is Popular?

- **High quality learning resources**
 - ▶ So many learning resources are available for new programmers.
- **Inbuilt functions and libraries**
 - ▶ When using Java, developers don't need to write every new function from scratch.
 - ▶ Java provides a rich ecosystem of **in-built functions and libraries to develop a range of applications.**

Why Java is Popular?

- **Active community support**

- ▶ Java has many active users and a community that can support developers when they **face coding challenges**.
- ▶ The Java platform software is also **maintained** and **updated** regularly.

- **High-quality development tools**

- ▶ Java offers various tools to support automated **editing, debugging, testing, deployment, and change management**.
- ▶ These tools make Java programming time and cost-efficient.

Why Java is Popular?

- **Platform Independent**

- ▶ Java code can run on any underlying platform like **Windows, Linux, or Android** without rewriting.
- ▶ This makes it especially powerful in today's environment, where we want to run applications on multiple devices.

- **Security**

- ▶ Users can download untrusted Java code over a network and run it in a secure environment in which it cannot do any harm.

Types of Java Applications

- **Standalone Application**

- ▶ Standalone applications are also known as **desktop applications** or **window-based applications**.



- **Web Application**

- ▶ An application that runs on the server side and creates a **dynamic page** is called a **web application**.

Types of Java Applications

- **Enterprise Application**

- ▶ An application that is distributed in nature, such as banking applications, etc. is called an enterprise application.

- **Mobile Application**

- ▶ An application which is created for mobile devices is called a mobile application.
- ▶ Currently, Android are used for creating mobile applications.

Java programming language used for

- Some common uses of Java include:
 - ▶ **Game Development**
 - ★ Many popular **mobile, computer, and video** games are **built in Java**.
 - ★ Even modern games that integrate advanced technology like machine learning or **virtual reality** are **built with Java technology**.
 - ▶ **Cloud computing**
 - ★ Java is often referred to as **WORA – Write Once and Run Anywhere**, making it perfect for decentralized **cloud-based applications**.
 - ▶ **Big Data**
 - ★ Java is used for data processing engines that can work with complex data sets and massive amounts of real-time data.

Java programming language used for

- Some common uses of Java include:
 - ▶ **Artificial Intelligence**
 - ★ Java is **powerhouse of machine learning libraries**.
 - ★ Its **stability and speed make it perfect** for artificial intelligence **application** development like **natural language processing and deep learning**.
 - ▶ **Internet of Things**
 - ★ Java has been used to program sensors and hardware in edge devices that can connect independently to the internet.

End of Chapter One

- Question
- Many Thanks!